

# ANN MARY ROY

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## EDUCATION

|                                                            |                     |
|------------------------------------------------------------|---------------------|
| <i>Cochin University of Science and Technology</i>         | July 2025- Present  |
| <b>M.Tech in VLSI &amp; Embedded Systems</b>               | <b>7.38 CGPA</b>    |
| <i>Government College of Engineering ,Kannur</i>           | Nov 2021- June 2025 |
| <b>B.Tech in Electronics and Communication Engineering</b> | <b>8.17 CGPA</b>    |
| (Minor: Computer Science and Engineering)                  |                     |
| <i>St. Mary's Convent School Alakode</i>                   | 2019 - 2020         |
| <b>Class 12 - CBSE</b>                                     | <b>95.2%</b>        |

## SKILLS

- **ASIC & FPGA Tools** : Cadence Virtuoso, Xcelium, Genus, Innovus, Tempus, Voltus, Conformal LEC, Vivado, Vitis
- **Programming Languages** : Verilog HDL, System Verilog, C, Embedded C, Python
- **Platforms** : STM32, ARM Cortex-M3, Zybo Z7, Raspberry Pi, Arduino

## INTERNSHIP EXPERIENCE

|                                                                |          |
|----------------------------------------------------------------|----------|
| <b>Student Intern - NIELIT ,Calicut</b>                        | Aug 2024 |
| • Worked on STM32 programming and peripheral-level debugging.  |          |
| <b>Internship Trainee - Keltron , Kannur</b>                   | May 2023 |
| • Exposure to electronics manufacturing and production systems |          |

## PROJECTS

|                                                                                                                                                                               |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>RTL to GDSII ASIC Design Flow Implementation</b>                                                                                                                           | Oct 2025 - Feb 2026 |
| • Implemented a complete <b>RTL-to-GDSII flow</b> , covering RTL coding, synthesis, and Physical Design (floorplanning, placement, CTS, routing).                             |                     |
| • Performed <b>Logical Equivalence Checking (LEC)</b> and <b>Static Timing Analysis (STA)</b> to ensure functional correctness and achieve timing closure.                    |                     |
| <b>RV32I Single-Cycle Processor</b>                                                                                                                                           | Dec 2025 - Jan 2026 |
| • Designed RV32I RISC-V processor including datapath, ALU, register file and control unit. Verified functionality using RTL simulation and modular testbench methodology.     |                     |
| <b>ASIC Verification, DFT &amp; Timing Analysis</b>                                                                                                                           | Nov 2025 - Mar 2026 |
| • Performed gate-level and SDF timing simulations using Cadence Xcelium (90 nm library). Implemented scan-based DFT and coverage-driven verification for sequential circuits. |                     |
| <b>FPGA SoC Design (Vivado/Vitis)</b>                                                                                                                                         | Jan 2026 - Mar 2026 |
| • Implemented embedded system design flows using Zynq architecture. Integrated AXI peripherals and performed synthesis and hardware validation.                               |                     |
| <b>Custom IC Design (Cadence Virtuoso)</b>                                                                                                                                    | Jul 2025 - Aug 2025 |
| • Designed CMOS inverter and logic gates, performed MOSFET characterization, CMOS circuit design layout implementation, and post-layout verification.                         |                     |
| <b>Lemon Sorting Robotic Arm</b>                                                                                                                                              | Aug 2024 - Mar 2025 |
| • Built automated sorting system using YOLOv8 + hardware integration (92% accuracy).                                                                                          |                     |
| <b>Plant Species Identification - Machine Learning Project</b>                                                                                                                | Aug 2024 - Jan 2025 |
| • Developed SVM model on Flavia dataset (1900+ images, 87% accuracy).                                                                                                         |                     |

## LEADERSHIP

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|---------------------------------------------------------------------------|--------------------|
| • University Union Councillor, M.Tech VLSI & Embedded Systems   CUSAT     | Sep 2025 - Present |
| • Malabar Hub Student Representative - IEEE IA/IE/PELS/Jt. Kerala Chapter | Feb 2024- Feb 2025 |

## CERTIFICATIONS

- Verilog SoC Design using Verilog HDL – Maven Silicon
- VLSI for Beginners – NIELIT
- Deep Learning – NPTEL